

WASP-43b

R-0568

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180416 · created 2026-07-24T14:37:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458224.7234 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.02
Transit depth (Rp/Rs)^2	2.03 +/- 0.57 [%]
Orbital Inclination (inc)	83.64 +/- 1.95
Airmass coefficient 1 (a1)	0.956 +/- 0.011
Airmass coefficient 2 (a2)	0.032 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#2 - [621, 328]
Optimal Aperture	4.32
Optimal Annulus	6.75
Transit Duration (day)	0.0535 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.02 vs 0.1594 (deviation 0.74σ)
Duration fit vs published	0.0535 d vs 0.0512 d (deviation 0.41σ)
Mid-time O–C	2.3 min
Depth SNR	6.4
Residual scatter	1.12 %

Light curve

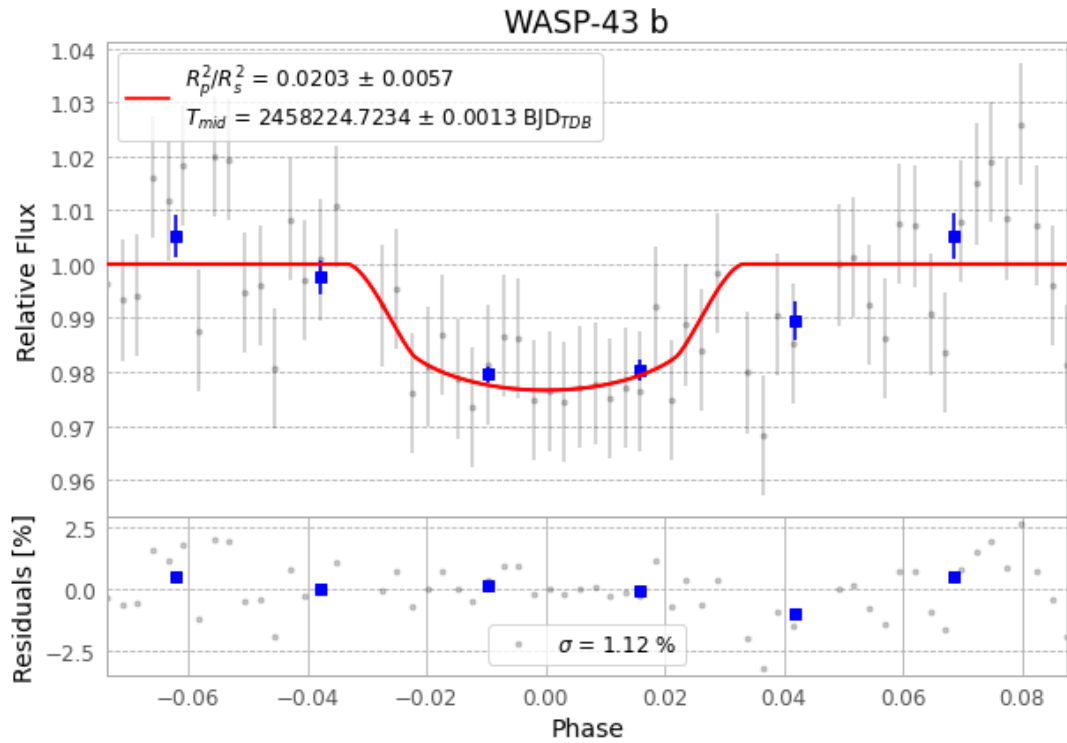


Figure 1 — FinalLightCurve_WASP-43 b_2018-04-15.png (SHA-256 55f1953830fb3e04...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6e3d1fd2e78359b4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

64 FITS frames (42 MB), WASP-43180416034812.FITS → WASP-43180416065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180416.log (4480bb0b0f57...), FinalLightCurve_WASP-43 b_2018-04-15.png (55f1953830fb...), FinalLightCurve_WASP-43 b_2018-04-15.csv (45a22f447d75...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 14:37Z.